

Bus a Dinner Table SOP

One episode buses one used table setting. The setting sits in the middle of the bare tabletop with a finished meal's tableware scattered loose across it. Nothing is stacked, nothing sits in a set position, the cutlery lies wherever it was put down, the paper napkin is crumpled, and the items sit at no fixed angle or spacing.

Every item is sorted into one of two receptacles, and **which receptacle an item belongs to is fixed by what the item is, never decided in the moment**. The **bus bin**, against the left edge, takes everything reusable: the dishes, the cutlery and the cup. The **waste bin**, against the right edge, takes the waste: the used paper napkin and the food scraps shaken off the dishes. Nothing else ever enters the waste bin, and no waste ever enters the bus bin.

Because the layout varies from episode to episode, **the clearing order is fixed by item type, not by position**: cup, then cutlery, then napkin, then dishes. That single rule is what keeps the data consistent across a scene that is different every time.

The reusable items are cleared in two passes, because **the order they have to come off the table is not the order they can go into the bus bin**. The tall cup and the loose cutlery have to leave the table first so nothing is lifted over them, but the dishes have to go into the bin first so they sit on the bottom as a stack. **Pass 1** strips every item off the tabletop, taking each dish to the waste bin for a shake on the way and the napkin to the waste bin for good. **Pass 2** loads the staged items into the bus bin in bin order: dishes, then cup, then cutlery.

The staging area is two separate places, and each item type has one of them. The **dish row** runs along the far edge of the tabletop, above the setting area. The **item column** is the strip of bare tabletop between the bus bin and the setting area, and it holds the cup and the cutlery. The napkin is never staged; it goes straight to the waste bin.

The two grippers split by side, and the two items that never change grippers split the same way. The **cup is a left gripper item from start to finish**, because the bus bin is on the left. The **paper napkin is a right gripper item from start to finish**, because the waste bin is on the right. Everything else crosses from the right gripper to the left gripper in a **handoff**, in mid air, so that the gripper that reaches the setting area is never the gripper that has to reach into the bus bin.

What stays constant across all episodes:

- **Waste and reusable:** the paper napkin and the food scraps go in the waste bin. The dishes, the cutlery and the cup go in the bus bin. Nothing crosses over.
- **Clearing order:** cup, then fork, knife, spoon, then napkin, then large plate, small plate, bowl.
- **Loading order:** large plate, small plate, bowl, then cup, then fork, knife, spoon.

- **Arm split:** the right gripper lifts items off the tabletop and out of the dish row, and works the waste bin. The left gripper takes the cup, receives every handoff, and lowers every item into the item column and into the bus bin.
- **Handoffs:** the cutlery crosses grippers in Pass 1. The plates and the bowl cross grippers in Pass 2. The cup and the napkin never cross grippers.
- **Bin trip:** every dish goes right to the waste bin and is tilted face down over it before it is staged, whether or not it is carrying scraps.
- **Dishes face up:** all three dishes are staged face up and stacked face up.
- **One item at a time:** exactly one item is lifted, carried and lowered per trip.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the bus bin on the left, the item column, the used setting in the middle, the dish row along the far edge, and the waste bin on the right.
3. Both arms are at home, low and clear of the tabletop, with grippers open.
4. The tabletop is dry.
5. Nothing obstructs the paths between the setting area, the item column, the dish row, the bus bin and the waste bin.
6. The left gripper can reach the floor of the bus bin and open there with the gripper clear of the bin's rim and walls.
7. The left gripper fits inside the mouth of the cup with room to open against the inner wall. If it does not fit, the inside grip is unavailable and the body grip is used for the whole session.
8. The right gripper can reach over the waste bin and open there, inside the line of the rim.
9. The space above the middle of the tabletop is clear, so the grippers can meet there for a handoff.

Materials checklist

1. Scattered loose on the bare tabletop, inside the setting area: a large plate, a small plate, a bowl, a fork, a knife, a soup spoon, a used paper napkin and a cup. No dish is stacked on

another dish.

2. The paper napkin is fresh for this episode, crumpled rather than folded, dry, and not torn into pieces. It is one napkin, not two stuck together.
3. The cup stands upright and is **empty and dry inside**, with its mouth clear, so either cup grip can be used. A cup with liquid or scraps in it is emptied before the episode starts.
4. The scatter is uneven, as a real finished meal leaves it: the fork and the knife on a plate, the spoon in the bowl, the napkin crumpled, items at no fixed angle.
5. Every item sits inside the setting area and within reach of the arm that owns it.
6. The three dishes carry loose, dry food scraps, or are empty. Either way each dish takes the bin trip in Step 4.
7. The bus bin is empty, dry, and sits against the left edge with its long edge running front to back. Its floor is clear.
8. The waste bin is empty or partly filled, and has clear room for the napkin and the scraps without anything sitting above its rim.
9. The dish row and the item column are clear.

Workspace layout

- **Bus bin:** on the tabletop against the left edge, within left gripper reach, long edge front to back. Takes the dishes, the cutlery and the cup.
- **Item column:** the strip of clear tabletop between the bus bin and the left edge of the setting area. Both grippers reach it. It holds the cup and the cutlery during Pass 1 and Pass 2, and is empty again at the end of the episode.
- **Dish row:** the strip of clear tabletop along the far edge, above the setting area. Both grippers reach it. It holds the three dishes during Pass 1 and Pass 2, and is empty again at the end of the episode.
- **Setting area:** the middle of the tabletop, between the item column and the waste bin, holding every item at the start of the episode.
- **Waste bin:** on the tabletop against the right edge, within right gripper reach. Takes the paper napkin and the shaken off scraps, and nothing else.

The right gripper works the setting area and the waste bin. The left gripper works the bus bin. Items cross between the grippers in a handoff above the middle of the tabletop, except the cup, which the left gripper carries on its own, the napkin, which the right gripper carries on its own, and the dishes in Pass 1, which the right gripper sets down in the dish row itself.

Vocabulary

Actions

- **Lift:** close the gripper on the item and raise it straight up until it is clear of everything around it.
- **Carry:** move a held item level, on a clear path, without touching anything else.
- **Lower:** bring a held item straight down until it rests on the tabletop, the floor of the bus bin or the stack, then open the gripper.
- **Bin:** carry the paper napkin to over the waste bin, lower it until the whole napkin is inside the line of the rim, and open the gripper. The napkin is the only item ever released this way, and it is the only item released without first resting on a surface.
- **Handoff:** the holding gripper stops and holds the item still. The receiving gripper closes on a free part of the same item. Both grippers hold the item for 1 second. Then the first gripper opens and moves clear. A handoff happens in mid air, above the middle of the tabletop, never over the waste bin and never over the bus bin.
- **Shake:** tilt a dish face down over the waste bin and give it two short shakes, then return it to level.
- **Stack:** lower a dish onto the dish below it, centred, so the stack sits square.

Arm assignments

- **The cup is a left gripper item from start to finish.** The left gripper lifts it off the tabletop in Step 1 and loads it into the bus bin in Step 6. The right gripper never touches it, and the cup is never handed off.
- **The paper napkin is a right gripper item from start to finish.** The right gripper lifts it off the tabletop in Step 3 and bins it in the same trip. The left gripper never touches it, and the napkin is never handed off and never staged.
- **The right gripper owns the setting area, the waste bin and the dish row.** It lifts every item except the cup off the tabletop, runs the bin trip, bins the napkin, sets the dishes down in the dish row, and lifts them back out of it. It never reaches over or into the bus bin.
- **The left gripper owns the item column and the bus bin.** It receives every handoff, lowers the cutlery into the item column, lifts it back out, and lowers every item into the bus bin. Apart from the cup it never reaches the setting area, and it never reaches toward the waste bin.
- **Cutlery, Pass 1:** the right gripper lifts the utensil off the tabletop and hands it to the left gripper. The left gripper lowers it into the item column.
- **Napkin:** the right gripper lifts it and carries it right to the waste bin in one trip. No handoff, no staging, no second pass.
- **Dishes, Pass 1:** the right gripper lifts the dish, carries it right to the waste bin, shakes it, then carries it left and lowers it face up in the dish row. No handoff.

- **Dishes, Pass 2:** the right gripper lifts the dish out of the dish row and hands it to the left gripper. The left gripper carries it to the bus bin and lowers it onto the stack.
- **Cutlery, Pass 2:** the left gripper lifts the utensil out of the item column and lowers it into the bus bin. No handoff.
- **Cup grip:** the cup may be lifted either way, and the left gripper picks one at each pickup:
 - **Body grip:** the gripper closes on the outside of the cup, around its body.
 - **Inside grip:** the gripper goes straight down into the mouth of the cup and opens against the inner wall until it holds. Both grips are equally correct. Use whichever suits the cup and what is around it; the inside grip is the one to reach for when the outside of the cup is not clear. Either way the cup stays upright from pickup to release and is never tipped to get a grip, and the grip is not changed while the cup is in the air.
- **Releasing an inside grip:** the release runs the other way round. Lower the cup until it stands on its own, **close** the gripper clear of the inner wall, then withdraw it straight up and out of the mouth. The cup must not lift, drag or rock as the gripper comes out. A body grip releases normally, by opening.
- **Bowl tilt:** the bowl may tilt during pickup or handover to give both grippers a usable grip angle. The left gripper levels it again before lowering it. The bowl is never turned over.
- Every item except the napkin is released only onto the tabletop, in the item column or the dish row, onto the floor of the bus bin or the stack inside it, or into the other gripper. The napkin is released inside the line of the waste bin. Never open a gripper over open space or above the bus bin's rim.

Working a scattered setting

- **Type order, not position order.** Clear cup, then cutlery, then napkin, then dishes, no matter where each item happens to be lying. Do not clear by whatever is nearest.
- Within the cutlery, the order is always **fork, knife, spoon**. Within the dishes, always **large plate, small plate, bowl**.
- **Never pull an item out from under another.** If the item due next is pinned or covered, lift the covering item straight up first.
- **Napkin branch:** if the napkin is draped over the cup or over a utensil, take the napkin first, bin it, then return to the type order from the top. This is the only permitted departure from the order.
- Take each grip where the item is clear, not where the standard grip would be on a tidy table: a plate by any free stretch of rim, a utensil across any free part of its handle.
- Never drag an item across the tabletop to get a better grip. Lift it straight up.
- If a grip fails, open the gripper, lift clear of the item, and try again. Two failed attempts on the same item is the limit before the episode is flagged.

Dish row positions

The dish row runs along the far edge of the tabletop, above the setting area. The three dishes sit face up in a row, side by side and never stacked. The row is filled from the bus bin side outward, in the order the dishes leave the setting area, so it reads **large plate, small plate, bowl** from the bin side across. That is also the order they will be loaded, so no dish is ever lifted over another.

Leave a clear gap between dishes so each one can be lifted straight up without brushing its neighbours.

Item column positions

The item column runs front to back on the strip of tabletop between the bus bin and the setting area. Each item has a fixed spot in it, and the column is filled from the far end forward:

- **Cup spot**, at the far end, level with the far corner of the bus bin: the cup standing upright.
- **Cutlery spots**, forward of the cup spot: the three utensils flat and parallel, handle ends roughly in line, heads toward the far edge, laid down in **fork, knife, spoon** order. They lie beside one another and never cross.

The near end of the column stays empty. Nothing in the item column touches, overlaps or rests on anything else, and nothing rests on the bus bin or against its rim.

Bus bin zones

Every item is lowered to the floor of the bus bin in an assigned zone, and every item ends up fully below the bin's rim:

- **Dish stack**: on the floor of the bin at the far end, largest at the bottom and all three face up: large plate, then small plate, then bowl.
- **Cup**: standing upright on the floor of the bin in the near half, in front of the stack, clear of the stack and of the cutlery.
- **Cutlery**: laid flat together on the floor of the bin in the near half, beside the cup and clear of the stack.

Nothing rests on the bin's rim, nothing is wedged against a wall, nothing overhangs the bin, and nothing sits on top of the dish stack. No waste goes into the bus bin.

Handling standard

- Move exactly one item at a time.
- Carry every item level and lower it straight down. There are two exceptions: a dish tilted face down over the waste bin to shake in Step 4, and a dish tilted for grip during a handoff in Step 5. A tilted dish is levelled again before it is lowered.

- Lower each item into the bus bin until it is resting on the bin's floor or on the stack, then release. Never open the gripper above the rim and let the item fall in.
- In a handoff, both grippers hold the item for 1 second before the first one opens.
- Nothing is slid. Correct a bad placement by lifting the item and setting it down again.
- An item is settled when it stays still for 2 seconds after release.

Steps

Step 1: Clear the cup to the item column

Goal: the cup stands upright at the cup spot and the tallest obstacle is off the tabletop.

- If the napkin is draped over the cup, bin the napkin first (see Step 3), then come back here.
- The **left gripper** takes the cup with either a **body grip** or an **inside grip** (see Cup grip under Arm assignments), lifts it straight up clear of whatever is around it, and carries it upright to the **cup spot**, the far end of the item column.
- Lower it straight down until it stands on its own, then release: open the gripper on a body grip, or close it clear of the inner wall and withdraw it straight up on an inside grip.
- This is the only time the left gripper reaches the setting area. It stays clear of the setting area for the rest of the episode, and the right gripper never touches the cup.

Check: the cup stands upright and stable at the cup spot, clear of the bus bin and of the rest of the column, and it did not move as the gripper came away. If it rocks or leans, lift it and set it down again.

Step 2: Clear the cutlery to the item column

Goal: the fork, knife and soup spoon lie in their spots in the item column.

- Move one utensil at a time in this order, wherever each one happens to be lying: **fork, knife, spoon.**
- If the utensil due next is pinned under the napkin or another utensil, lift the covering item straight up and set it clear on the tabletop, then take the utensil in order. If the covering item is the napkin, bin it rather than setting it back down.
- If a utensil is resting on a dish or standing in the bowl, lift it straight up and out. Do not slide it off.
- The **right gripper** grips the utensil across a free section of its handle and lifts it clear.
- The **right gripper** hands the utensil to the **left gripper**.
- The **left gripper** lowers the utensil flat into the item column, forward of the cup.
- Lay the three side by side in the order they arrive, heads toward the far edge, handle ends roughly in line. Keep a gap between them and do not lay one across another.

- Keep every utensil clear of the bus bin. A utensil that lands against the bin's rim is lifted and set down again.

Check: all three utensils lie flat and parallel in the item column, clear of the cup and of the bus bin. Each one changed grippers in mid air. Nothing was dragged across the tabletop to get to them.

Step 3: Bin the paper napkin

Goal: the used paper napkin is inside the waste bin, and no part of it is left on the table.

The napkin is waste. It never goes to the item column, never changes grippers, and never goes near the bus bin. The **right gripper** does the whole step in one trip.

- The **right gripper** pinches the napkin at a raised fold or a free edge and lifts it clear of the table.
- Before carrying it, check that nothing came up with it. If a utensil, a scrap or part of a dish is caught in the napkin, set the napkin back down, free the item, and lift the napkin again.

Never carry the napkin to the waste bin with anything caught in it.

- The **right gripper** carries the napkin right, to over the waste bin.
- **Bin it:** lower the napkin until the whole of it is inside the line of the rim, then open the gripper and release it.
- If the napkin tears and part of it stays on the table, bin the piece in hand first, then lift and bin each remaining piece the same way. Every piece goes in the waste bin.

Check: the whole napkin is inside the waste bin, no piece of it is left in the setting area or on the tabletop, nothing else went into the bin with it, and it never touched the item column or the bus bin.

Step 4: Shake and stage the three dishes

Goal: the large plate, small plate and bowl have each been shaken over the waste bin and sit face up in the dish row.

Work one dish at a time in this order, wherever each one is lying: **large plate, then small plate, then bowl.** The **right gripper** does the whole step. There is no handoff in this step, and the left gripper stays clear.

4.1 Lift and carry to the waste bin

- The **right gripper** grips the dish by any free stretch of rim and lifts it level off the tabletop.
- It carries the dish right, to over the waste bin.

4.2 Shake it over the waste bin

- Tilt the dish face down toward the bin, with the whole rim inside the line of the bin.
- Give it **two short shakes**, then return the dish to level. Whatever comes off, comes off.

- Shake every dish, including an empty one.

4.3 Carry it across to the dish row

- The **right gripper** returns the dish to level, carries it left above the setting area, and lowers it face up in the **dish row** along the far edge.
- Fill the row from the bus bin side outward, so each dish lands further from the bin than the one before: the **large plate** nearest the bin, the **small plate** beside it, the **bowl** at the outer end.
- The three sit side by side and never stacked, with a clear gap between them.

Check: all three dishes are face up in the dish row, separate, and reading large plate, small plate, bowl from the bus bin side outward. Each was tilted and shaken over the waste bin on its way there. The setting area is bare.

Expected state: the dish row reads large plate, small plate, bowl from the bus bin side outward. The item column reads, from the far end forward, the upright cup, the fork, the knife, then the spoon. The setting area is bare and the tabletop around it is clear. The waste bin holds the napkin and the shaken off scraps. The bus bin is still empty.

Step 5: Load the dish stack into the bus bin

Goal: the three dishes are stacked face up on the floor of the bus bin, largest at the bottom.

Load in this order, which is the dish row worked from the bin side outward, so no dish is lifted over another: **large plate, then small plate, then bowl.**

5.1 Load the two plates

- The **right gripper** grips the plate by its rim and lifts it level out of the dish row.
- The **right gripper** hands the plate to the **left gripper**.
- The **left gripper** carries the plate to the bus bin, lowers it in over the rim, and sets it down face up: the large plate onto the floor of the bin at the far end, and the small plate onto the large plate.
- Centre each plate on the one below it so the stack sits square rather than leaning, then release.

5.2 Load the bowl

- The **right gripper** grips the bowl by its rim, lifts it out of the dish row, and hands it to the **left gripper**.
- The bowl may tilt during pickup or handover to give both grippers a usable grip angle.
- The **left gripper** carries the bowl to the bus bin and levels it out.
- The **left gripper** lowers the bowl face up onto the small plate, centred, and releases once it rests still.
- The bowl is never turned over and never rests face down.

Check: the stack reads large plate, small plate, bowl from the bottom up, all three are face up, and the whole stack sits below the bin's rim. If the stack leans or a dish rocks, lift that dish and lower it again. Do not shove it into place.

Step 6: Load the cup

Goal: the cup stands upright on the floor of the bus bin in front of the stack.

- The **left gripper** takes the cup at the cup spot with either a **body grip** or an **inside grip** (see Cup grip under Arm assignments) and lifts it straight up. The grip does not have to match the one used in Step 1.
- It carries the cup upright to the bus bin and lowers it in over the rim onto the floor of the bin, in the near half in front of the stack, clear of the stack and of the cutlery zone.
- The cup does not change grippers at any point.
- Release it only once it stands upright and still on the floor of the bin: open the gripper on a body grip, or close it clear of the inner wall and withdraw it straight up on an inside grip. Bring the gripper up and out of the bin without disturbing the cup.
- The cup is never laid on its side, never wedged against a wall of the bin, never left propped on the rim, and never goes on top of the stack.

Check: the cup stands upright on the floor of the bus bin, fully below the rim, touching nothing else, and it did not move as the gripper came away.

Step 7: Load the cutlery

Goal: the bus bin is fully loaded and the item column is empty.

- Load in this order: **fork, knife, spoon**. That is the column worked from the far end forward, so nothing is lifted over another item.
- The **left gripper** lifts one utensil at a time from the item column and lays each flat on the floor of the bin in the near half, beside the cup. The three lie together and parallel.
- There is no handoff in this step. The left gripper works the item column on its own.
- All three utensils are loaded. A utensil that has been lifted is only ever released onto the floor of the bus bin. It is never carried right and never released over the waste bin.

Check: the item column and the dish row are both empty. All three utensils rest on the floor of the bus bin beside the cup. Nothing rests on the rim and nothing overhangs the bin.

Step 8: End the episode

Goal: the table is clear, the waste is binned and both arms are safely home.

- Confirm the setting area, the item column and the dish row are all bare.
- Confirm the bus bin holds the three dish stack, the three utensils and the upright cup, and the waste bin holds only the paper napkin and the shaken off scraps.

- Return both arms home, low and clear of the tabletop, with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Unload the bus bin.
2. Empty the waste bin: take out the used napkin and the scraps. Confirm no reusable item is in there before emptying it.
3. Scatter the tableware again in the setting area on the bare tabletop: the three dishes loose and unstacked, the three utensils loose with at least one resting on a dish or in the bowl, and the cup upright. Vary the scatter between episodes rather than rebuilding the same arrangement.
4. Add a **fresh paper napkin**, crumpled rather than folded, and vary where it lands between episodes: sometimes clear of everything, sometimes draped over the cup or a utensil so the napkin branch gets exercised. The napkin is consumed every episode and is never reused.
5. Put loose, dry scraps back on the three dishes.
6. Confirm every item is inside the setting area and within reach of the arm that owns it.
7. Wipe the tabletop. Clear the item column and the dish row.
8. Clean or replace anything chipped, cracked, bent or stained.
9. Put the empty bus bin back against the left edge and the waste bin against the right edge.
10. Run both Setup checklists again.

SOP violations

These are operator actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every operator violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because an operator broke a rule.

Violations

Violation: Waste sorted into the bus bin

- **Visible cue:** the paper napkin is put into the bus bin, staged in the item column, or handed to the left gripper at any point; or scraps are dumped into the bus bin.
- **SOP rule broken:** Step 3 (the waste bin takes the paper napkin and the scraps; no waste ever enters the bus bin; the napkin is a right gripper item from start to finish).
- **Coaching note:** the napkin is rubbish. It only ever goes right.

Violation: Reusable item sorted into the waste bin

- **Visible cue:** a utensil, the cup or a dish is released into the waste bin, or is left resting in it.
- **SOP rule broken:** Steps 4 to 7 (the bus bin takes the dishes, the cutlery and the cup; the waste bin receives only the napkin and the scraps).
- **Coaching note:** only the napkin and the scraps go in the waste bin. Everything else comes back.

Violation: Napkin binned with something caught in it

- **Visible cue:** the napkin is lifted with a utensil, a scrap-covered dish or another item tangled in it and is carried to the waste bin anyway, instead of being set back down and freed first. Also counts when the caught item is released into the waste bin with the napkin.
- **SOP rule broken:** Step 3 (check that nothing came up with the napkin; set it back down, free the item, and lift again; move exactly one item at a time).
- **Coaching note:** look at what you are holding before you carry it right. A utensil in the bin is a lost utensil.

Violation: Napkin released outside the waste bin

- **Visible cue:** the gripper opens while the napkin is not fully inside the line of the rim, and the napkin lands on the tabletop, on the bin's outside, or on the floor. Also counts when a torn piece of the napkin is left on the table.
- **SOP rule broken:** Step 3 (lower the napkin until the whole of it is inside the line of the rim before opening the gripper; every torn piece goes in the waste bin).
- **Coaching note:** get the whole napkin over the opening, take it down, then let go.

Violation: Cleared by position instead of by type

- **Visible cue:** items come off the tabletop in whatever order they happen to be reachable, such as a dish before the cutlery, the napkin before the cup, or the cutlery split around a dish, rather than cup, then all cutlery, then napkin, then dishes.
- **SOP rule broken:** Steps 1 to 4 (clear by type order: cup, cutlery, napkin, dishes; no matter where each item is lying, do not clear by whatever is nearest).
- **Coaching note:** the scatter changes every episode; the order never does.

Violation: Wrong order within a type

- **Visible cue:** the utensils leave the tabletop in any order other than fork, knife, spoon, or the dishes leave in any order other than large plate, small plate, bowl.
- **SOP rule broken:** Steps 2 and 4 (cutlery always fork, knife, spoon; dishes always large plate, small plate, bowl).
- **Coaching note:** name the next item before you reach for it.

Violation: Item pulled out from under another

- **Visible cue:** a utensil is dragged out from beneath the napkin or another utensil, a dish is pulled out from under something resting on it, or an item is slid sideways to free it, instead of the covering item being lifted straight up first.
- **SOP rule broken:** Step 2 (never pull an item out from under another; lift the covering item straight up first, then take the pinned item in type order).
- **Coaching note:** uncover it, then lift it. Never tug it free.

Violation: Napkin branch misused

- **Visible cue:** the napkin is taken out of order when it was not covering the cup or a utensil, or the napkin is left in place and the cup or a utensil is pulled out from under it, or the napkin is lifted off a covered item and set back down on the table instead of being binned.
- **SOP rule broken:** Steps 1, 2 and 3 (take the napkin first only when it is draped over the cup or a utensil, bin it, then return to the type order from the top).
- **Coaching note:** the napkin jumps the queue only when it is covering something, and once it is up it goes straight in the bin.

Violation: Handoff skipped

- **Visible cue:** the right gripper sets a utensil down in the item column itself, or the left gripper reaches the setting area to take one; or the left gripper lifts a dish out of the dish row itself instead of receiving it from the right gripper. The item never changes grippers.
- **SOP rule broken:** Steps 2 and 5 (the cutlery crosses grippers when it is cleared; the plates and the bowl cross grippers when they are loaded).
- **Coaching note:** the gripper that reaches the setting area is never the gripper that reaches into the bin. Hand it over in the middle.

Violation: Handoff done wrong

- **Visible cue:** the first gripper opens before the second has closed on the item, or both open together and the item falls, or the two grippers hold the item for less than 1 second, or the item changes grippers over the waste bin, over the bus bin, or off the edge of the tabletop.
- **SOP rule broken:** Steps 2 and 5 (both grippers hold the item for 1 second; a handoff happens in mid air above the middle of the tabletop, never over either bin).
- **Coaching note:** both grippers on the item, count one second, then let go.

Violation: Item handed off that should not be

- **Visible cue:** the cup or the napkin changes grippers at any point, or a dish is handed off in Step 4 instead of being staged by the right gripper, or a utensil is handed off in Step 7 instead of being loaded by the left gripper.
- **SOP rule broken:** Steps 1, 3, 4, 6 and 7 (the cup is a left gripper item and the napkin a right gripper item, both from start to finish; the right gripper stages the dishes in Step 4 on its own; the left gripper loads the cutlery in Step 7 on its own).
- **Coaching note:** handoffs belong to the cutlery going out and the dishes coming back. Nothing else.

Violation: Item staged in the wrong place

- **Visible cue:** a staged item is not in its assigned area or spot: a dish anywhere but the dish row, or stacked on another dish; a utensil or the cup anywhere but the item column. Also counts when an area reads in the wrong order, such as the bowl nearest the bus bin or a utensil at the far end of the column ahead of the cup.
- **SOP rule broken:** Steps 1 to 4 (dishes face up in the dish row on the far edge, reading large plate, small plate, bowl from the bus bin side outward; the cup at the far end of the item column, then the cutlery forward of it).
- **Coaching note:** put each item in its own spot as it arrives. Staging is not a pile.

Violation: Staged item left against the bus bin

- **Visible cue:** a staged utensil or the cup comes to rest touching the bus bin, leaning on its rim, or partly underneath it, and is left that way.
- **SOP rule broken:** Step 2 (nothing staged in the item column rests on the bus bin or against its rim).
- **Coaching note:** set it down in the open column, then check the gap to the bin before releasing.

Violation: Staged item lifted over another

- **Visible cue:** during Steps 5 to 7 a gripper reaches across or over a staged item to reach one behind it, most often taking the bowl before the plates or the spoon before the fork.
- **SOP rule broken:** Steps 5 to 7 (work the dish row from the bus bin side outward and the item column from the far end forward; nothing is lifted over another staged item).
- **Coaching note:** always take the next item in the row or column. If you are reaching over something, the order is wrong.

Violation: Wrong arm used for an item

- **Visible cue:** the right gripper picks up or loads the cup; the left gripper touches the napkin; the left gripper lifts anything other than the cup off the tabletop in the setting area; the left

gripper reaches toward the waste bin; the right gripper reaches over or into the bus bin.

- **SOP rule broken:** Steps 1 to 7 (the cup is a left gripper item and the napkin a right gripper item, both from start to finish; the right gripper owns the setting area, the waste bin and the dish row; the left gripper owns the item column and the bus bin).
- **Coaching note:** the cup is always the left gripper's, the napkin always the right gripper's. Only the left gripper puts anything in the bus bin.

Violation: Dish not shaken over the waste bin

- **Visible cue:** a dish goes from the setting area to the dish row without being carried right to the waste bin and tilted face down, or it is tilted somewhere other than over the waste bin.
- **SOP rule broken:** Steps 4.1 and 4.2 (the right gripper carries each dish to the waste bin and tilts it face down, with the whole rim inside the line of the bin, before the dish is staged).
- **Coaching note:** every dish goes right before it goes left, including an empty one.

Violation: Scraps shaken outside the waste bin

- **Visible cue:** a dish is tilted and shaken while it is over the tabletop, the setting area, the dish row or the bus bin, and scraps land outside the waste bin. Also counts when a dish is tilted on the carry back across to the dish row.
- **SOP rule broken:** Steps 4.1, 4.2 and 4.3 (tilt only while the whole rim is over the waste bin; return the dish to level before carrying it across).
- **Coaching note:** level it before you leave the bin.

Violation: Item missing from the bus bin at the end

- **Visible cue:** the episode reaches Step 8 with a dish, a utensil or the cup not in the bus bin, including an item that was lifted and then released somewhere other than the bin.
- **SOP rule broken:** Steps 5 to 8 (every staged item is loaded into the bus bin; the final check confirms the three dish stack, the three utensils and the cup).
- **Coaching note:** count the load before you home the arms: three dishes, three utensils, cup.

Violation: Wrong bin load order

- **Visible cue:** the dish stack is built in any order other than large plate, small plate, bowl, or the cup or the cutlery goes into the bin before the stack is complete, or the cutlery goes in before the cup.
- **SOP rule broken:** Steps 5 to 7 (the dish stack goes in first, largest to smallest; then the cup; then the cutlery).
- **Coaching note:** the stack big to small, then the cup, then the cutlery.

Violation: More than one item moved at once

- **Visible cue:** two utensils, two dishes, or a dish with a utensil still on it are lifted together.

- **SOP rule broken:** Steps 1 to 7 (move exactly one item at a time).
- **Coaching note:** one item per trip, every trip.

Violation: Item dropped into the bus bin instead of lowered

- **Visible cue:** a gripper opens above the bus bin's rim and the item falls the rest of the way, instead of being lowered in until it rests on the floor of the bin or on the stack.
- **SOP rule broken:** Steps 5 to 7 (lower each item into the bin until it is resting, then release; never open the gripper above the rim and let the item fall in). Releasing the napkin inside the line of the waste bin is not a violation.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Dish stack built incorrectly

- **Visible cue:** the stack in the bus bin is out of size order, any of the three dishes is face down, a dish is clearly off centre on the one below it, or the stack leans and is left that way.
- **SOP rule broken:** Step 5 (large plate, small plate, bowl, all face up, each centred on the one below it; lift any dish and lower it again that leaves the stack leaning).
- **Coaching note:** all three face up, and square each dish on the one under it before releasing.

Violation: Bowl carried or stacked face down

- **Visible cue:** the bowl is turned over during pickup, handover or carry, or it comes to rest face down on the stack. A tilt for grip that is levelled again before lowering is not a violation.
- **SOP rule broken:** Step 5.2 (the bowl may tilt for grip, is levelled by the left gripper before lowering, and is never turned over).
- **Coaching note:** tilt it if you must to get the grip, then bring it back level before it lands.

Violation: Cup left lying down or not standing in the bin

- **Visible cue:** the cup comes to rest on its side, leaning, on top of the stack, wedged against a wall of the bus bin, or propped on the rim, and is left that way.
- **SOP rule broken:** Step 6 (the cup stands upright on the floor of the bin in front of the stack, fully below the rim, touching nothing else).
- **Coaching note:** hold the cup until it stands on its own, then let go.

Violation: Loose item stacked on the dishes

- **Visible cue:** a utensil or the cup ends up resting on top of the dish stack rather than in its own zone on the floor of the bus bin.
- **SOP rule broken:** Steps 6 and 7 (cup upright in front of the stack, cutlery flat beside the cup, both clear of the stack).
- **Coaching note:** the stack is not a shelf. Every loose item goes on the bin floor.

Violation: Item left resting on the bin rim or overhanging

- **Visible cue:** the episode ends with any loaded item touching the bus bin's rim, propped across it, or sticking out past the outside of the bin, instead of sitting fully below the rim.
- **SOP rule broken:** Steps 5 to 8 (every item ends up fully below the bin's rim; nothing rests on the rim and nothing overhangs the bin).
- **Coaching note:** if you can see it above the rim line, it is not loaded yet.

Violation: Item slid instead of lifted

- **Visible cue:** a gripper pushes, nudges or drags an item across the tabletop or across the floor of the bus bin, either to reposition it for a better grip or to correct a placement, instead of lifting it clear and lowering it again. The item stays in contact with the surface while it moves.
- **SOP rule broken:** Steps 1 to 7 (carry every item level and lower it straight down; never drag an item across the tabletop to get a grip; correct a bad placement by lifting the item and setting it down again).
- **Coaching note:** lift it clear and put it down again. Nothing gets shoved into position.

Violation: Repeated re-grips on one item

- **Visible cue:** a gripper closes on the same item and fails to lift it more than twice, or opens and re-closes on an item already held more than twice, before the item moves.
- **SOP rule broken:** Steps 1 to 7 (if a grip fails, open the gripper, lift clear, and try again; two failed attempts on the same item is the limit).
- **Coaching note:** reset the approach after the second miss instead of fishing for the item.

Violation: Staging left occupied

- **Visible cue:** the episode reaches Step 8 with an item still sitting in the item column or the dish row, or an item is left behind in the setting area.
- **SOP rule broken:** Steps 7 and 8 (the item column, the dish row and the setting area are all bare before the episode ends).
- **Coaching note:** sweep the column and the row with your eyes before homing.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an arm away from home or a gripper closed, or recording stops before both arms are home.
- **SOP rule broken:** Step 8 (confirm the bare setting area, item column and dish row, the loaded bus bin and the binned waste; return both arms home with grippers open; then stop recording).
- **Coaching note:** run the final confirmation, then home, then stop.

Failures that are not violations

These failures are not the operator's fault. Log them as system issues, discard the episode, and never use them to coach the operator.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Reach fault on a handoff:** the two arms cannot meet above the middle of the tabletop because of a rig or calibration fault rather than an operator choice.
- **Reach fault into the bus bin:** the left gripper cannot reach the floor of the bin, or cannot open there without fouling the rim or a wall, because of a rig or calibration fault.
- **Napkin tears under a correct grip:** the napkin comes apart on a normal pinch and lift rather than from tugging it out from under something. Bin every piece, then log the episode and replace the napkin stock if it keeps happening.
- **Defective object:** a chipped, cracked or bent item found mid episode, or a bus bin or waste bin that shifts off its zone. Replace it before the next episode.

Annotation subtasks

1. Move the cup from the setting area to the item column
2. Uncover a pinned item by lifting what is on top of it
3. Lift one utensil off the tabletop
4. Hand one item from the right gripper to the left gripper
5. Lower one utensil into the item column
6. Lift the paper napkin off the tabletop
7. Carry the paper napkin to the waste bin and release it inside the rim
8. Carry one dish from the setting area to the waste bin
9. Shake one dish over the waste bin
10. Carry one shaken dish across to the dish row
11. Lift one dish out of the dish row
12. Stack one dish face up in the bus bin
13. Load the cup upright into the bus bin
14. Load one utensil into the bus bin
15. Return both arms home and end the episode